COLE RUOFF

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Education

The University of Texas MD Anderson Cancer Center UTHealth Houston Graduate School of Biomedical Sciences (GSBS) August 2024 – Present
Houston, Texas
- Doctor of Philosophy - PhD in Quantitative Sciences and Cancer Biology

University of Maryland August 2018 – May 2022
College Park, Maryland
- Bachelor of Science - B.S. in Computer Science

Research Experience

Graduate Research Assistant May 2025 – Present
MD Anderson Cancer Center
Advisor: Nicholas Navin

- Investigated tumor evolution, therapy resistance, and metastasis in breast and prostate cancer via single-cell RNA and DNA sequencing data
- Performed tissue dissociation and single-cell sequencing experiments on prostate and lymph node cancer biopsies
- Collaborated with clinicians to coordinate tumor specimen collection for clinical studies

Rotation Student January 2025 – April 2025
MD Anderson Cancer Center
Advisor: Loukia Karacosta

- Analyzed single-cell proteomics data to study treatment responses in small-cell lung cancer
- Processed blood liquid biopsies and prepared samples for cytometry by time-of-flight (CyTOF)
- Co-first authored manuscript detailing findings in proteomics data

Post-baccalaureate Fellow June 2022 – August 2024
National Institutes of Health
Advisor: Sridhar Hannenhalli

- Analyzed single-cell RNA sequencing data to identify non-genetic mechanisms of therapeutic resistance in various cancer cell lines
- Identified malignant cell states harboring resistance mechanisms in early drug response data
- Developed computational biology skills and familiarized myself with various bioinformatics tools through conducting research for individual project
- Drafted and compiled first-author research article detailing the findings of my research
- Presented individual research at multiple conferences and poster sessions

Research Intern February 2021 – June 2022
National Institutes of Health
Advisor: Sridhar Hannenhalli

- Developed a machine learning approach to identifying the presence of a target cell type in low resolution spatial transcriptomics data
- Consulted and collaborated with multiple immunology laboratories to identify T cell functional activity in cancer microenvironment spatial transcriptomics data
- Implemented an unsupervised breadth-first expansion algorithm relying on coexpression of genes to identify novel cell states in single-cell RNA sequencing data

Research Assistant May 2020 – August 2020
University of Maryland
Advisor: Brantley Hall

- Executed high-performance computing jobs with sequencing data from genomes of microorganisms from the human microbiome
- Developed Python scripts to analyze and annotate sequencing data (e.g. promoter discovery, motif identification, etc.)

Undergraduate Researcher January 2019 – December 2020
University of Maryland
Advisor: Catherine Spirito

- Led a research team to develop a biosensor that can detect trace amount of microRNA biomarkers for cancer
- Performed experimental procedures including PCR, Gel electrophoresis, DNA/RNA purification and quantification

Publications

1. **Ruoff, C.,** Mitchell, A., Mondal, P., Gopalan, V., Singh, A., Gottesman, M., & Hannenhalli, S. (2025). *Resistance signatures manifested in early drug response across cancer types and species*. **Cancer Drug Resistance**, 8, 44. https://doi.org/10.20517/cdr.2025.112

2. Wang, K., Taslic, C. A., de Groot, P., Parma, M. A., Paula, A. G., Caranto, M., Shah, K., Bose, M., **Ruoff, C**., Karacosta, L. G., Byers, L. A., Gay, C. M., Zhang, J., Heymach, J. V., & Zhang, B. (2025). *First Report of Response to Tarlatamab in a Patient With Histologic-Transformed SCLC From ALK-Rearranged NSCLC*: Case Report. **JTO clinical and research reports,** 7(2), 100941. https://doi.org/10.1016/j.jtocrr.2025.100941

Presentations

1. **Ruoff C,** Xu P, Zhao Y, Yan Y, Ye R, Tang C, Sei E, Bai S, Tran T, Ly C, Zhu K, Zhang M, A. Siddiqui B, K. Subudhi S, Corn P, Logothetis C, Troncoso P, J. Zurita A, Navin N (10/2026). *Delineating Genomic Features of Heterogeneity and Clinical Progression in Metastatic Prostate Cancer.* **Quantitative Sciences Seminar.** Oral Presentation
2. **Ruoff C,** Xu P, Zhao Y, Yan Y, Ye R, Tang C, Sei E, Bai S, Tran T, Ly C, Zhu K, Zhang M, A. Siddiqui B, K. Subudhi S, Corn P, Logothetis C, Troncoso P, J. Zurita A, Navin N (09/2026). *Delineating Genomic Features of Heterogeneity and Clinical Progression in Metastatic Prostate Cancer.* **Cancer Biology Program Retreat.** Poster
3. **Ruoff C,** Xu P, Zhao Y, Yan Y, Ye R, Tang C, Sei E, Bai S, Tran T, Ly C, Zhu K, Zhang M, A. Siddiqui B, K. Subudhi S, Corn P, Logothetis C, Troncoso P, J. Zurita A, Navin N (07/2026). *Genomic Heterogeneity Shapes Metastatic Phenotype and Clinical Outcome in Prostate Cancer.* **GSBS Graduate Student Research Day.** Poster
4. **Ruoff C,** Xu P, Zhao Y, Yan Y, Ye R, Tang C, Sei E, Bai S, Tran T, Ly C, Zhu K, Zhang M, A. Siddiqui B, K. Subudhi S, Corn P, Logothetis C, Troncoso P, J. Zurita A, Navin N (04/2026). *Genetically Diverse Metastatic Sites Converge Transcriptionally in Prostate Cancer.* **Quantitative Sciences Program Retreat.** Poster
5. **Ruoff C**, Bose M, Stewart A, Byers L, F Lujan E, Ehsan E, Karacosta L (05/2025). *Phenotyping circulating tumor cells in small-cell lung cancer liquid biopsies via mass cytometry.* **Cancer Systems Imaging Department Retreat.** Poster
6. **Ruoff C**, Bose M, Stewart A, Byers L, F Lujan E, Ehsan E, Karacosta L (04/2025).

Phenotyping circulating tumor cells in small-cell lung cancer liquid biopsies via mass cytometry. **Quantitative Sciences Program Retreat.** Poster

7. **Ruoff C**, Wang J, Lin Y, Navin N (12/2024). *Investigating Drug Resistance in Triple-Negative Breast Cancer Using Single-cell DNA and RNA Data.* **Genetics & Epigenetics Rotation Blitz**, Oral Presentation
8. **Ruoff C**, Gopalan V, Hannenhalli S (05/2023). *Transcriptional Characterization of Drug-Resistant Cancer Cells*. **Center for Cancer Research Fellows and Young Investigators Colloquium**, National Cancer Institute, Rockville, MD. Poster
9. **Ruoff C**, Gopalan V, Hannenhalli S (12/2022). *Transcriptional Characterization of Drug-Resistant Cancer Cells*. **National Cancer Institute Cancer Data Science Laboratory Seminar Series**, National Institutes of Health, Bethesda, MD. Oral Presentation
10. **Ruoff C**, Gopalan V, Hannenhalli S (04/2022). *Combinatorial Approach to Detecting Emerging Rare Cell States in Tumors*. **University of Maryland Undergraduate Research Day**, University of Maryland, College Park, MD. Poster
11. **Ruoff C**, Gopalan V, Hannenhalli S (08/2021). *Delineating Cellular Composition in Spatial Transcriptomic Data*. **National Institutes of Health Summer Research Presentation Week**, National Institutes of Health, Bethesda, MD. Poster
12. **Ruoff C**, Ternovskaia C, Wirt E (11/2019). *DNA-Based Biosensor for MicroRNA Detection*. **University of Maryland First-year Innovation & Research Experience Summit**, University of Maryland, College Park, MD. Poster
13. **Ruoff C,** Wirt E, Barry L, Cao V, Goodson J, Winkler W, Spirito C (10/2019). *Aptamer Selection Against Mutant Bacterial Antiterminator Protein NasR R340K*. **University of Maryland, Baltimore County Undergraduate Research Symposium**, University of Maryland Baltimore County, Baltimore, MD. Poster

Leadership Activities

Genetics & Epigenetics Data Science Working Group Committee March 2025 - Present
GSBS

Quantitative Sciences Student Council Member May 2025 - Present
GSBS

Quantitative Sciences Steering Committee Student Representative Aug 2025 – Sept 2026
GSBS

Teaching & Mentoring

GCC Single Cell and Spatial Genomics Workshop Teaching Assistant May 2026

- Contributed to setting up computational resources for week-long workshop
- Answered questions from participants on tissue preparation, sequencing experiments, data analysis, and data visualization.

10X Genomics/Association of Science Communication Single-cell RNA Sequencing Analysis Workshop Teaching Assistant May 2026

- Assisted computational instruction during workshop
- Guided participants through specific analysis steps and helped troubleshoot analysis issues

Gulf Coast Consortium Fellows Program October 2025 - Present

- Served as a mentor to undergraduate students in the Gulf Coast Consortium Scholars

Program
- Provided feedback and guidance on the graduate school application process

Graduate Student Association Peer Mentor September 2025 – Present
- Assisted first-year students in navigating the Texas Medical Center
- Answered general questions from first-year students involving graduate school milestones

University of Maryland FIRE Program Peer Mentor January 2020 – December 2020
- Trained new laboratory students in technical skills including pipette usage, PCR, gel electrophoresis, DNA/RNA purification and quantification, etc.
- Instructed students on general biology knowledge along with concepts involving aptamers, biosensors, and sequencing

Awards

2026 IDSO & JPAI Colin Stroud Hackathon 1st place May 2026

2026 QS Retreat Pre-Candidacy Poster Competition 3rd place April 2026

GSBS Endowment Scholarship December 2025

National Cancer Institute's Cancer Research Training Award June 2022